Linux Lab Report

file system and access control management lab

April 20, 2026

Part 1 – Files & Directories

Commands Executed

Create the following directory structure :

```
mia@fedora:~$ sudo mkdir -p /home/lab/company/{hr,it,finance}
mia@fedora:~$
```

Inside each directory: Create 2 empty files. Create one file with content

```
mia@fedora:~$ sudo touch /home/lab/company/hr/file1.txt /home/lab/company/hr/file2.txt
mia@fedora:~$ sudo touch /home/lab/company/it/file1.txt /home/lab/company/it/file2.txt
mia@fedora:~$ sudo touch /home/lab/company/finance/file1.txt /home/lab/company/finance/file2.txt
mia@fedora:~$ sudo echo "hr data" > /home/lab/company/hr/hr_info.txt
bash: /home/lab/company/hr/hr_info.txt: Permission denied
mia@fedora:~$ echo "hr data" | sudo tee /home/lab/company/hr/hr_info.txt
hr data
mia@fedora:~$ cat /home/lab/company/hr/hr_info.txt
hr data
mia@fedora:~$ echo "it data" | sudo tee /home/lab/company/it/it_info.txt
it data
mia@fedora:~$ cat /home/lab/company/it/it_info.txt
it data
mia@fedora:~$ echo "finance data" | sudo tee /home/lab/company/finance/finance_info.txt
finance data
mia@fedora:~$ cat /home/lab/company/finance/finance_info.txt
finance data
mia@fedora:~$
```

Show detailed info about the created structure

```
mia@fedora:~$ ls -R /home/lab/company
/home/lab/company:
finance  hr  it

/home/lab/company/finance:
file1.txt  file2.txt  finance_info.txt

/home/lab/company/hr:
file1.txt  file2.txt  hr_info.txt

/home/lab/company/it:
file1.txt  file2.txt  it_info.txt
mia@fedora:~$
```

Part 2 – Links

Commands Executed

In /home/lab/company/hr, create a file employees.txt

```
mia@fedora:~$ echo "list of employees" | sudo tee /home/lab/company/hr/employees.txt
[sudo] password for mia:
list of employees
mia@fedora:~$
```

Create: A hard link → employees_hard.txt A symbolic link → employees_soft.txt

```
mia@fedora:~$ sudo ln /home/lab/company/hr/employees.txt /home/lab/company/hr/employees_hard.txt
mia@fedora:~$ sudo ln -s /home/lab/company/hr/employees.txt /home/lab/company/hr/employees_soft.txt
mia@fedora:~$
```

Delete the original file and observe behavior

```
mia@fedora:~$ ls -li /home/lab/company/hr/
total 16
21049 -rw-r--r--. 2 root root 18 Apr 20 10:13 employees_hard.txt
21050 lrwxrwxrwx. 1 root root 34 Apr 20 10:15 employees_soft.txt -> /home/lab/company/hr/employees.txt
21049 -rw-r--r--. 2 root root 18 Apr 20 10:13 employees.txt
21035 -rw-r--r--. 1 root root  0 Apr 20 09:57 file1.txt
21036 -rw-r--r--. 1 root root  0 Apr 20 09:57 file2.txt
21044 -rw-r--r--. 1 root root  8 Apr 20 10:03 hr_info.txt
mia@fedora:~$ sudo rm /home/lab/company/hr/employees.txt
mia@fedora:~$
```

What happens to each link after deletion? Why?

```
mia@fedora:~$ cat /home/lab/company/hr/employees_hard.txt
list of employees
mia@fedora:~$ ls -li /home/lab/company/hr/employees_hard.txt
21049 -rw-r--r--. 1 root root 18 Apr 20 10:13 /home/lab/company/hr/employees_hard.txt
mia@fedora:~$
```

```
mia@fedora:~$ cat /home/lab/company/hr/employees_soft.txt
cat: /home/lab/company/hr/employees_soft.txt: No such file or directory
mia@fedora:~$ ls -li /home/lab/company/hr/employees_soft.txt
21050 lrwxrwxrwx. 1 root root 34 Apr 20 10:15 /home/lab/company/hr/employees_soft.txt -> /home/lab/compa
ny/hr/employees.txt
mia@fedora:~$
```

hard links survive deletion of the original because they share the same inode.
symbolic links break because they depend on the file path, not the data itself.

Part 3 – Users & Groups

Commands Executed

Create the three following groups: hr_team, it_team, finance_team

```
mia@fedora:~$ sudo groupadd hr_team
[sudo] password for mia:
mia@fedora:~$ sudo groupadd it_team
mia@fedora:~$ sudo groupadd finance_team
mia@fedora:~$
```

Create the following users: alice (HR employee), bob (IT employee), charlie (Finance employee)

```
mia@fedora:~$ sudo useradd alice
mia@fedora:~$ sudo useradd bob
mia@fedora:~$ sudo useradd charlie
mia@fedora:~$
```

Assign each user to their respective group

```
mia@fedora:~$ sudo usermod -aG hr_team alice
mia@fedora:~$ sudo usermod -aG it_team bob
mia@fedora:~$ sudo usermod -aG finance_team charlie
mia@fedora:~$
```

Create an additional shared group: management

```
mia@fedora:~$ sudo groupadd management
mia@fedora:~$
```

Add alice and bob to management

```
mia@fedora:~$ sudo usermod -aG management alice
mia@fedora:~$ sudo usermod -aG management bob
mia@fedora:~$
```

What is the difference between primary and secondary groups?

```
mia@fedora:~$ id alice
uid=1009(alice) gid=1015(alice) groups=1015(alice),1012(hr_team),1018(management)
mia@fedora:~$ id bob
uid=1010(bob) gid=1016(bob) groups=1016(bob),1013(it_team),1018(management)
mia@fedora:~$ id charlie
uid=1011(charlie) gid=1017(charlie) groups=1017(charlie),1014(finance_team)
mia@fedora:~$
```

the default group gets assigned to a user when they are created and it gets stored in /etc/passwd as the GID. every user has exactly one primary group, and when a user creates a file, that file is automatically owned by their primary group.
but a user can belong to multiple secondary groups as well but they have to be added manually.
it gets stored in /etc/groups and it is used to give users access to shared resources.

Part 4 – Permissions & ownership

Commands Executed

Apply permissions: Only HR group can access /hr, Only IT group can access /it, Finance directory is read-only
Set: Owner: respective department user, Group: respective department group

```
mia@fedora:~$ sudo chown alice:hr_team /home/lab/company/hr
[sudo] password for mia:
mia@fedora:~$ sudo chown bob:it_team /home/lab/company/it
mia@fedora:~$ sudo chown charlie:finance_team /home/lab/company/finance
mia@fedora:~$ sudo chmod g+r,g-wx,o-rwx /home/lab/company/finance
mia@fedora:~$ sudo chmod g+rwx,o-rwx /home/lab/company/hr
mia@fedora:~$ sudo chmod g+rwx,o-rwx /home/lab/company/it
mia@fedora:~$
mia@fedora:~$ ls -l /home/lab/company/
total 0
drwxr-----. 1 charlie finance_team  68 Apr 20 10:06 finance
drwxrwx---. 1 alice   hr_team      130 Apr 20 10:16 hr
drwxrwx---. 1 bob     it_team       58 Apr 20 10:05 it
mia@fedora:~$
```

Test access: Login as different users, Try accessing other directories

```
mia@fedora:~$ sudo passwd alice
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
mia@fedora:~$ su - alice
Password:
alice@fedora:~$ ls /home/lab/company/hr/
employees_hard.txt  employees_soft.txt  file1.txt  file2.txt  hr_info.txt
alice@fedora:~$ ls /hoome/lab/company/it
ls: cannot access '/hoome/lab/company/it': No such file or directory
alice@fedora:~$ exit
logout
mia@fedora:~$
mia@fedora:~$ sudo passwd bob
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
mia@fedora:~$ su - bob
Password:
bob@fedora:~$ ls /home/lab/company/it/
file1.txt  file2.txt  it_info.txt
bob@fedora:~$ ls /home/lab/company/hr/
ls: cannot open directory '/home/lab/company/hr/': Permission denied
bob@fedora:~$ exit
logout
mia@fedora:~$
```

Give management group read access to all directories

```
mia@fedora:~$ sudo chown -R alice:management /home/lab/company/hr
[sudo] password for mia:
mia@fedora:~$ sudo chown -R bob:management /home/lab/company/it
mia@fedora:~$ sudo chown -R charlie:management /home/lab/company/finance
mia@fedora:~$ sudo chmod g+rx /home/lab/company/hr
mia@fedora:~$ sudo chmod g+rx /home/lab/company/it
mia@fedora:~$ sudo chmod g+rx /home/lab/company/finance
mia@fedora:~$
```

```
mia@fedora:~$ ls -l /home/lab/company/
total 0
drwxr-x---. 1 charlie management  68 Apr 20 10:06 finance
drwxrwx---. 1 alice   management 130 Apr 20 10:16 hr
drwxrwx---. 1 bob     management  58 Apr 20 10:05 it
mia@fedora:~$
```

```
mia@fedora:~$ su - bob
Password:
bob@fedora:~$ ls /home/lab/company/hr
employees_hard.txt  employees_soft.txt  file1.txt  file2.txt  hr_info.txt
bob@fedora:~$ ls /home/lab/company/finance/
file1.txt  file2.txt  finance_info.txt
bob@fedora:~$ exit
logout
mia@fedora:~$
```

Part 5 – Access Control

Commands Executed

A confidential file must: Be readable only by alice. Be editable by management. Be invisible to others.
Create it and apply the correct: Ownership, Group, Permissions

```
mia@fedora:~$ sudo touch /home/lab/company/confidential.txt
[sudo] password for mia:
mia@fedora:~$ echo "confidential data" > /home/lab/company/confidential.txt
bash: /home/lab/company/confidential.txt: Permission denied
mia@fedora:~$ echo "confidential data" | sudo tee /home/lab/company/confidential.txt
confidential data
mia@fedora:~$ sudo chown alice:management /home/lab/company/confidential.txt
mia@fedora:~$ sudo chmod u-wx,g+rw,o-rwx /home/lab/company/confidential.txt
mia@fedora:~$ ls -l /home/lab/company/confidential.txt
-r--rw----. 1 alice management 18 Apr 20 11:21 /home/lab/company/confidential.txt
mia@fedora:~$ su - alice && ca
cache_check          cache_writeback      canberra-gtk-play    cardos-tool
cache_dump           cal                  cancel               case
cache_metadata_size  ca-legacy            cancel.cups          cat
cache_repair         caller               capsh                catman
cache_restore        canberra-boot        captoinfo
mia@fedora:~$ su - alice && cat /home/lab/company/confidential.txt  && exit
Password:
```

```
mia@fedora:~$ su - alice && cat /home/lab/company/confidential.txt  && exit
Password:
su: Authentication failure
mia@fedora:~$ su - alice | cat /home/lab/company/confidential.txt |  exit
cat: /home/lab/company/confidential.txt: Permission denied
Password:
Broken pipe
mia@fedora:~$ su - alice
Password:
su: Authentication failure
mia@fedora:~$ su - alice
Password:
alice@fedora:~$ cat /home/lab/company/confidential.txt
confidential data
alice@fedora:~$ exit
logout
```

```
mia@fedora:~$ su - charlie
Password:
charlie@fedora:~$ cat /home/lab/company/confidential.txt
cat: /home/lab/company/confidential.txt: Permission denied
charlie@fedora:~$ exit
logout
mia@fedora:~$
```